

Machine learning for data analytics

Assignment 4: Time Series II & Deep Learning

Janusz Szwabiński

Each assignment should be submitted as a Jupyter notebook with complete code, visualizations, and written analysis.

LSTM for Multivariate Forecasting (30 points)

Build an LSTM model for energy consumption prediction:

- Use a multivariate dataset (temperature, humidity, day of week, energy consumption).
- Implement sequence-to-one and sequence-to-sequence architectures.
- Experiment with different sequence lengths and hidden units.
- Compare with classical methods (ARIMA, Prophet).

Attention Mechanisms (25 points)

Implement attention layer for sequence modeling:

- Add attention mechanism to LSTM model from Problem 4.1.
- Visualize attention weights over input sequences.
- Analyze which time steps receive highest attention.

Temporal Fusion Transformer (25 points)

Use PyTorch Forecasting library:

- Apply TFT to a retail sales forecasting problem.
- Include static features (store location) and time-varying features (promotions, holidays).
- Analyze variable importance from the model.
- Generate probabilistic forecasts with quantile predictions.

Model Comparison (20 points)

Comprehensive evaluation of time series models:

- Compare ARIMA, Prophet, LSTM, and TFT on same dataset.
- Evaluate using RMSE, MAE, and MAPE.
- Analyze computational costs.
- Provide recommendations for different use cases.